

How ML Really Works?

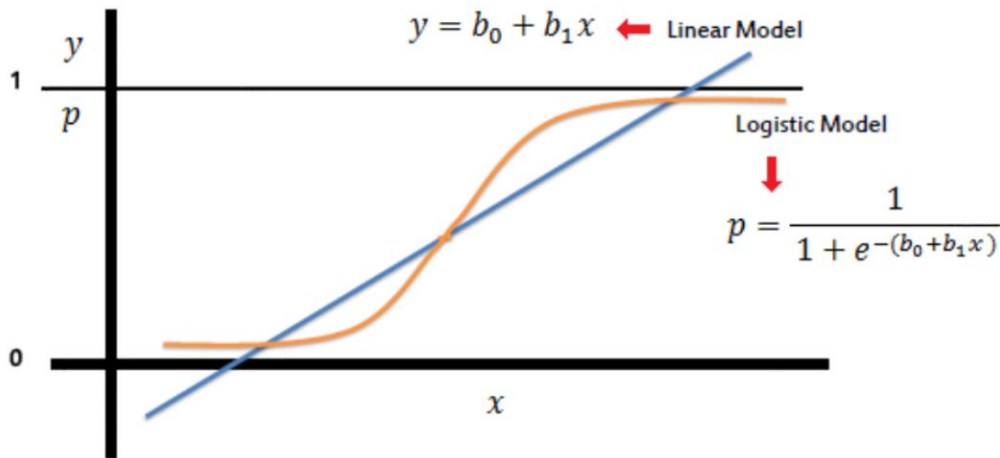
Logistic Regression

Logistic regression is a supervised learning classification algorithm used to predict the **probability** of a target variable. The nature of target or dependent variable is dichotomous, which means there would be **only two possible classes**.

In simple words, the dependent variable is binary in nature having data coded as either **1** (stands for success/yes) or **0** (stands for failure/no).

Logistic Regression Assumptions

- In case of binary logistic regression, the target variables must be **binary** always and the desired outcome is represented by the factor level 1.
- There should **not** be any **multi-collinearity** in the model, which means the independent variables must be independent of each other .
- We must include **meaningful variables** in our model.
- We should choose a **large sample size** for logistic regression.



Logistic Regression uses the **sigmoid function**, was developed by statisticians to describe properties of population growth in ecology, rising quickly and maxing out at the carrying capacity of the environment. It's an **S-shaped curve** that can take any real-valued number and map it into a value **between 0 and 1**, but never exactly at those limits.

The Sigmoid Function

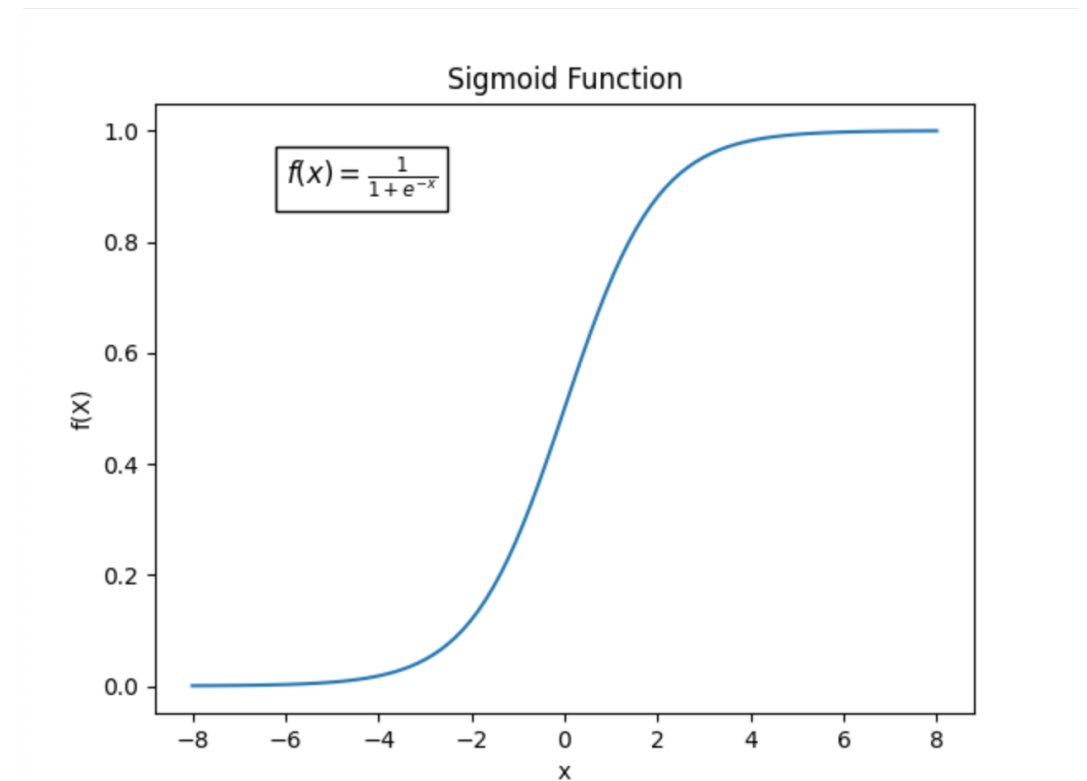
Sigmoid is a mathematical function that maps any real-valued number into a value **between 0 and 1**. Its characteristic "S"-shaped curve makes it particularly useful in scenarios where we need to convert **outputs into probabilities**. This function is often called the logistic function. Mathematically, sigmoid is represented as:

$$\sigma = \frac{1}{1 + e^{-x}}$$

where,
 x is the input value,
 e is Euler's number (≈ 2.718)

Explanation

- x is any real number — it could be -1000 , 0 , 7.5 , or anything else.
- e is Euler's number, approximately 2.718 . It is a fundamental constant in mathematics, like π .
- e^{-x} means "e raised to the power of negative x ."
- The formula wraps that exponential inside a fraction and adds 1 to the bottom.
- The result is always a number strictly between 0 and 1 , no matter what x is.



Logistic Regression - Python Implementation

Example:

Create a Machine Learning model that predicts whether an employee would have bought an SUV car. The Independent features are User ID, Gender, Age and EstimatedSalary. The dependent label is Purchased(1 for Yes, 0 for No).

```
1 import numpy as nm
2 import matplotlib.pyplot as plt
3 import pandas as pd
4 from sklearn.linear_model import LogisticRegression
5 from sklearn.preprocessing import StandardScaler
6 from sklearn.model_selection import train_test_split
7 from sklearn.metrics import confusion_matrix
8 from sklearn.metrics import accuracy_score
9
10 data_set= pd.read_csv('./sk_learn/suv_data.csv')
11
12 x= data_set.iloc[:, [2,3]].values
13 y= data_set.iloc[:, 4].values
14 X_train, X_test, y_train, y_test = train_test_split(x, y, test_size = 0.2, random_state = 1)
15
16 st_x= StandardScaler()
17 x_train= st_x.fit_transform(X_train)
18 x_test= st_x.transform(X_test)
19 classifier= LogisticRegression(random_state=0)
20 classifier.fit(x_train, y_train)
21
22 y_pred= classifier.predict(x_test)
23 cm= confusion_matrix(y_test, y_pred)
24 print(cm)
25 print("Accuracy: ",accuracy_score(y_test, y_pred))
```

Steps

1. Import Logistic Regression from **Sklearn** library.
2. Split data into train and test set.
3. **Standardize** the data.
4. Create a **Logistic Regression object**.
5. Train the model using fit().
6. **Evaluate** on test set.

Real-World Applications of Logistic Regression

1. Customer Churn Prediction

Predict whether a customer is likely to leave a service based on their usage, plan, and engagement.



2. Spam Email Detection

Classify emails as spam or legitimate based on features such as keywords, sender information, and email patterns.



3. Loan Default Prediction

Estimate whether a borrower is likely to default based on income, credit history, and loan details.



4. Disease Risk Prediction

Predict whether a patient is likely to have a particular disease based on symptoms, test results, and medical factors.



5. Ad Click Prediction

Predict whether a user is likely to click an advertisement based on their demographics, behavior, and browsing activity.



Resources to Learn More

Real Python : Logistic Regression implementation in Python, with detailed coding examples and intuition.

Articlesledge : Clear explanation of the **Sigmoid Function** and its role in Logistic Regression.

Google for Developers : Logistic Regression concepts, classification, probabilities, and model evaluation.

StatQuest : Excellent visual intuition for Logistic Regression, probability, and classification.

YouTube Channels

StatQuest with Josh Starmer : Excellent for intuition and mathematical understanding.

3Blue1Brown : Great for developing mathematical intuition around linear algebra and regression.

CampusX : Strong Hindi explanations; their 100 Days of ML series includes Logistic Regression intuition, mathematical formulation, coding, and regression metrics.